An Introduction to Control and Automation

Learning Objectives

- What will you know/be able to do by the end?
  - Explain the basic concept of control systems and automation.
  - Differentiate between open-loop and closed-loop control.
  - Understand the basics of PID control
  - Apply both ON/OFF and PID-based control in a simulated or real system.

Outline

Part 1: Basics of Control System

- Control and Automation
- Automation classifications: Flexible and Programmable
- Control System Types: Open-loop Vs. Closed-loop (Feedback) Control
- PID Control and Tuning

- Part 2: PLC Controller
  - PLC Basics and Ladder Programming

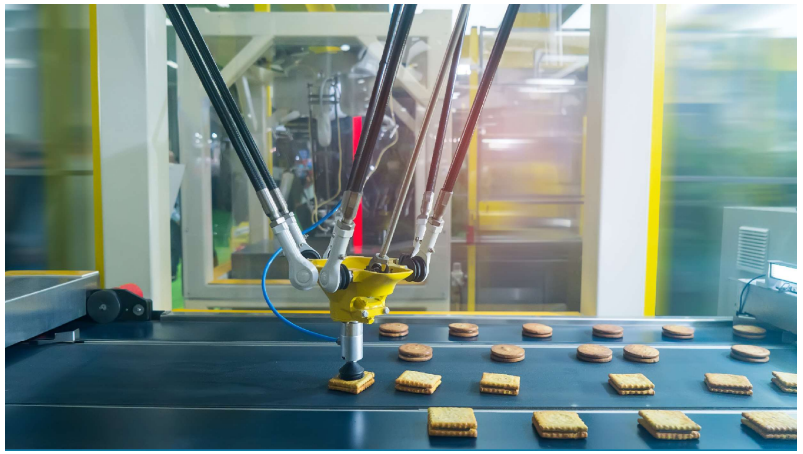
What is Automation and Control ?

Automation is the use of control systems and technologies to operate processes with **minimal human intervention**.

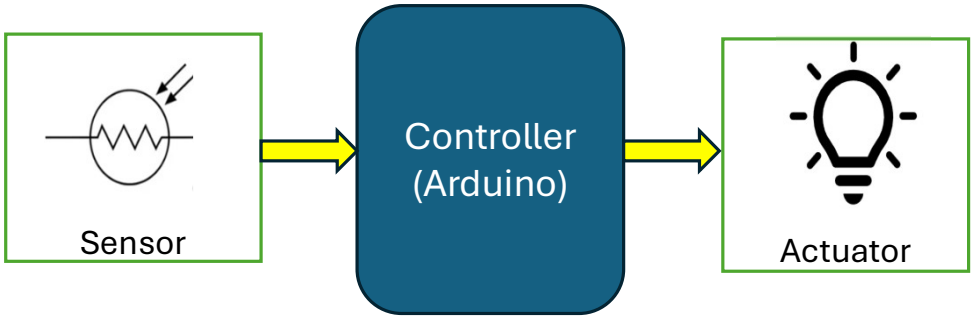
- Examples: manufacturing robots, CNC machines, automatic irrigation.

Control means regulating the behavior of a system to achieve desired output.

- Example: Maintaining room temperature using an air conditioner.

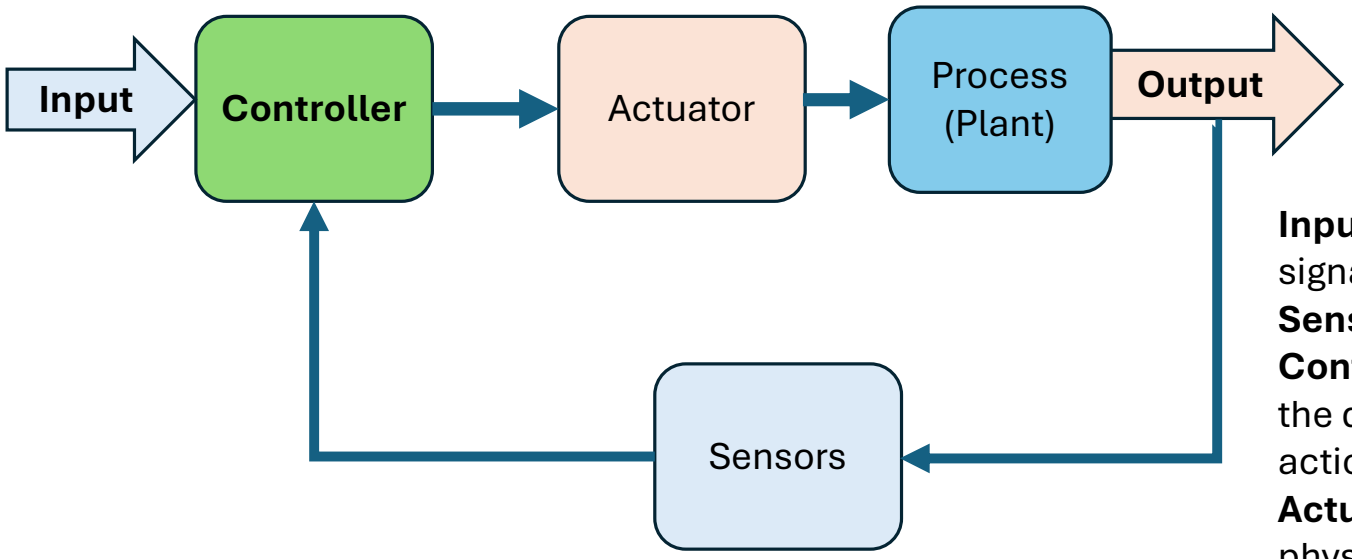


Components of a Control System





Components of a Control System



Input: The desired condition or command signal.
Sensor: Measures the actual condition
Controller : Compares measured value with the desired setpoint and decides the control action.
Actuator: Converts the control signal into physical action
Process: The system being controlled
Output: The final result or behavior (response) of the system or effect

Example: Temperature Control System

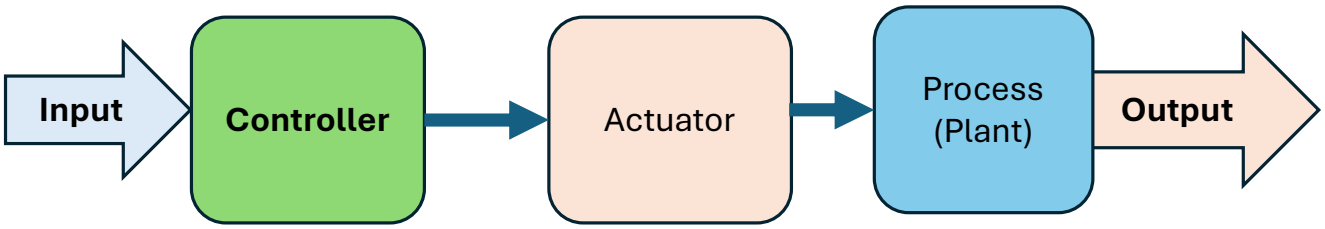
Element	Role	Example: Temperature Control System
Input (Setpoint)	Desired temperature	e.g., 70 °F
Actuator	Device that adds or removes heat	e.g., heater, fan, or cooling coil
Sensor	Measures current temperature	e.g., thermistor or thermocouple
Controller	Compares desired vs actual and decides how much to heat or cool	e.g., Arduino, PID controller, or thermostat circuit
Process / Plant	The physical environment whose temperature is being controlled	The room, furnace chamber, incubator, or oven

Control Types

1. Open Loop Control (No-feedback loop)
2. Closed Loop Control (Feedback Control)

Open-loop Control

- Open-loop systems do **not use feedback,** they execute the command regardless of the output result

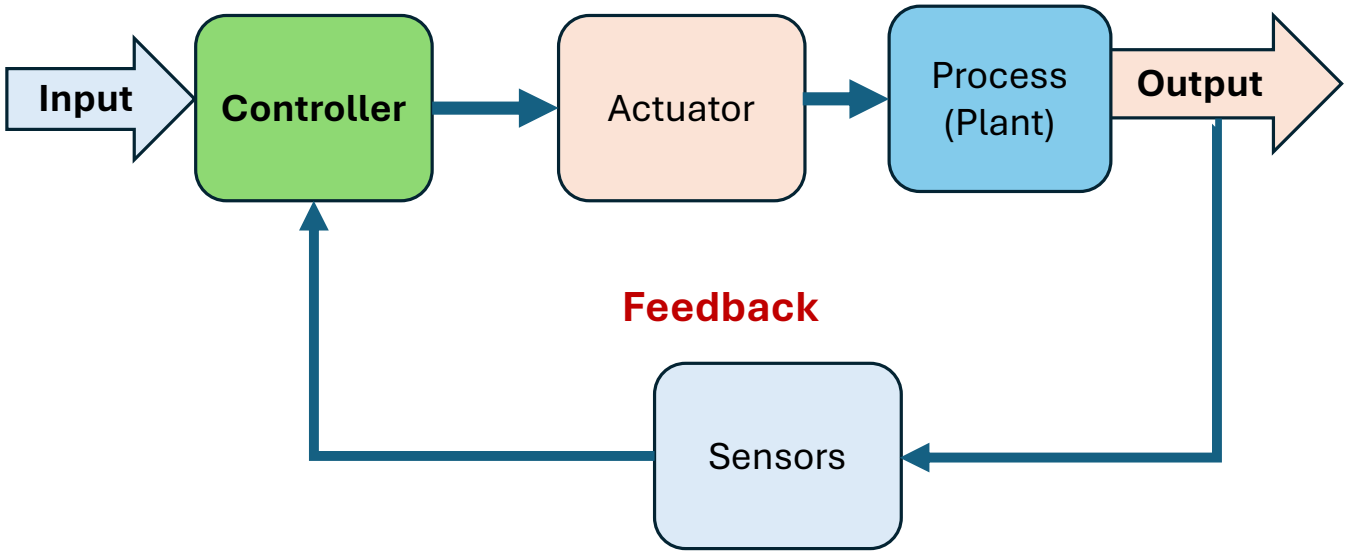


Examples:
Microwave Oven:
It doesn't sense the actual temperature of the food.

Sprinkler Timer System:

It doesn't check soil moisture or rain conditions.

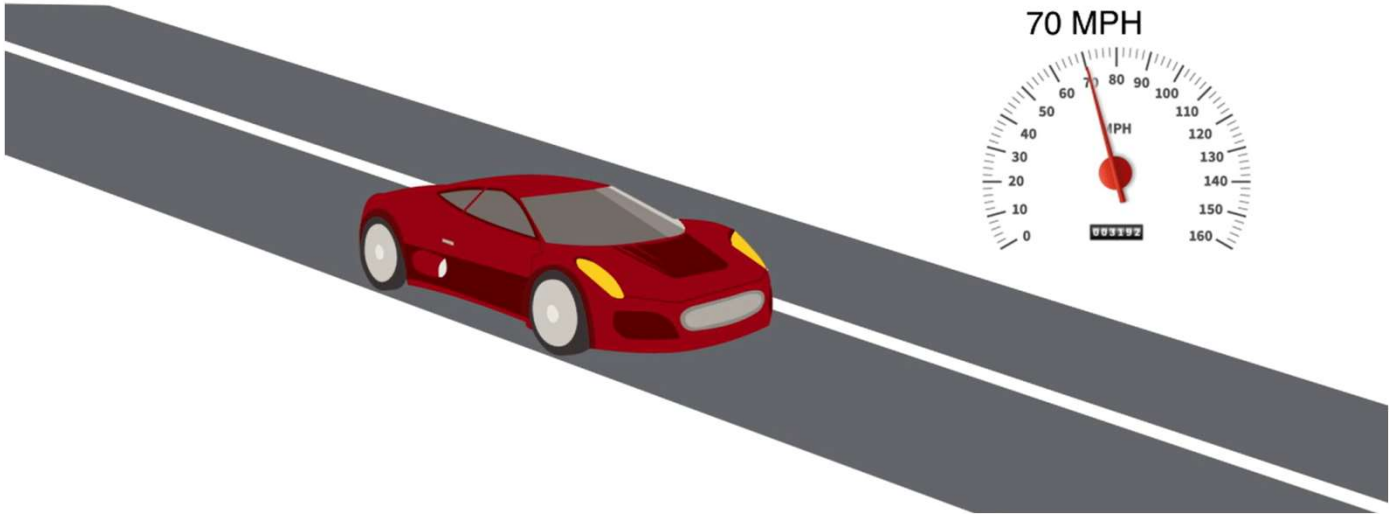
Closed-loop (Feedback) Control

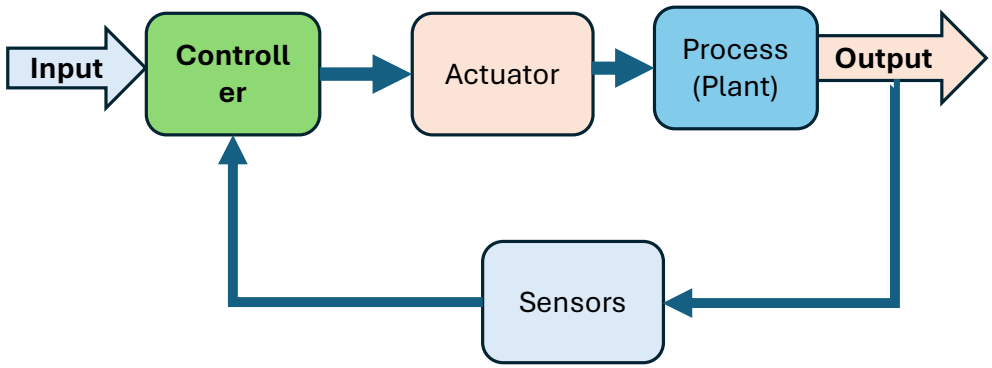


Closed-loop systems use **feedback loop** (sensors inputs) to compare actual output with the desired value and automatically adjust.

Close-loop Control Example- Cruise Control

Goal: Maintains a set speed by adjusting throttle despite disturbances like hills or wind resistance.



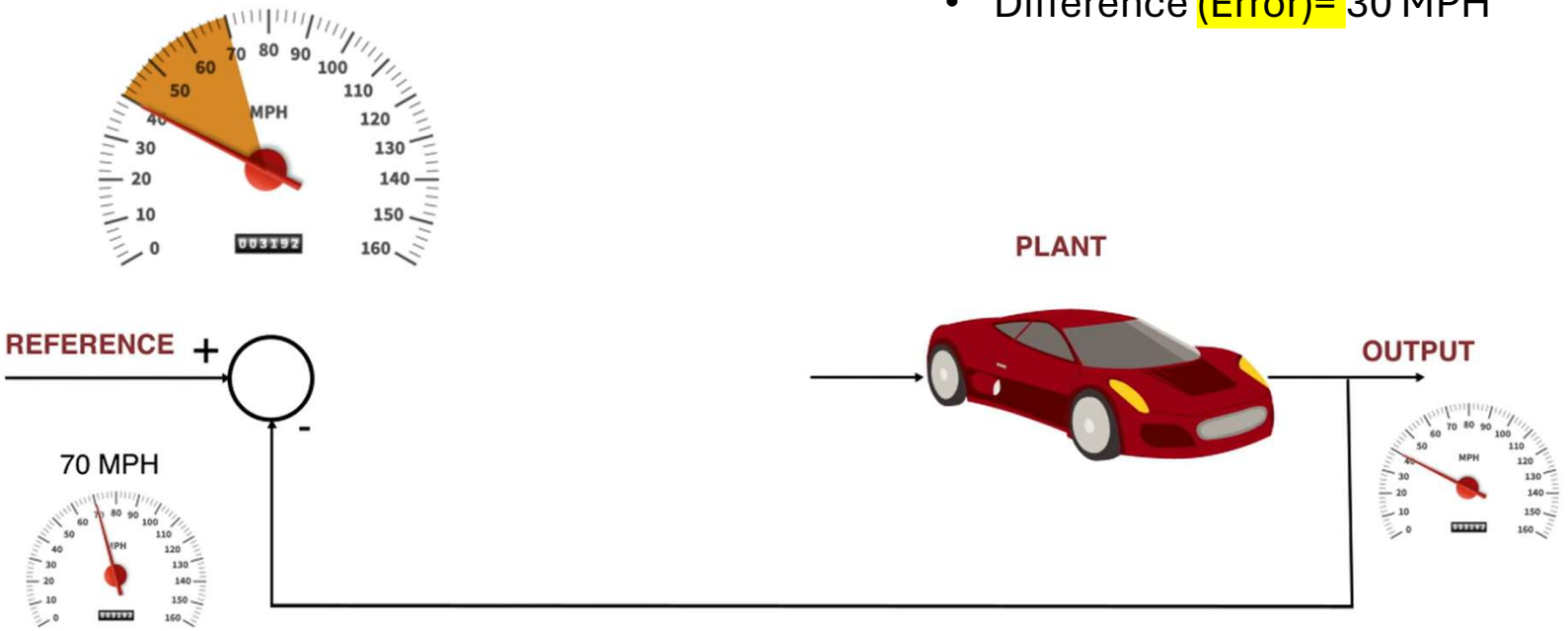


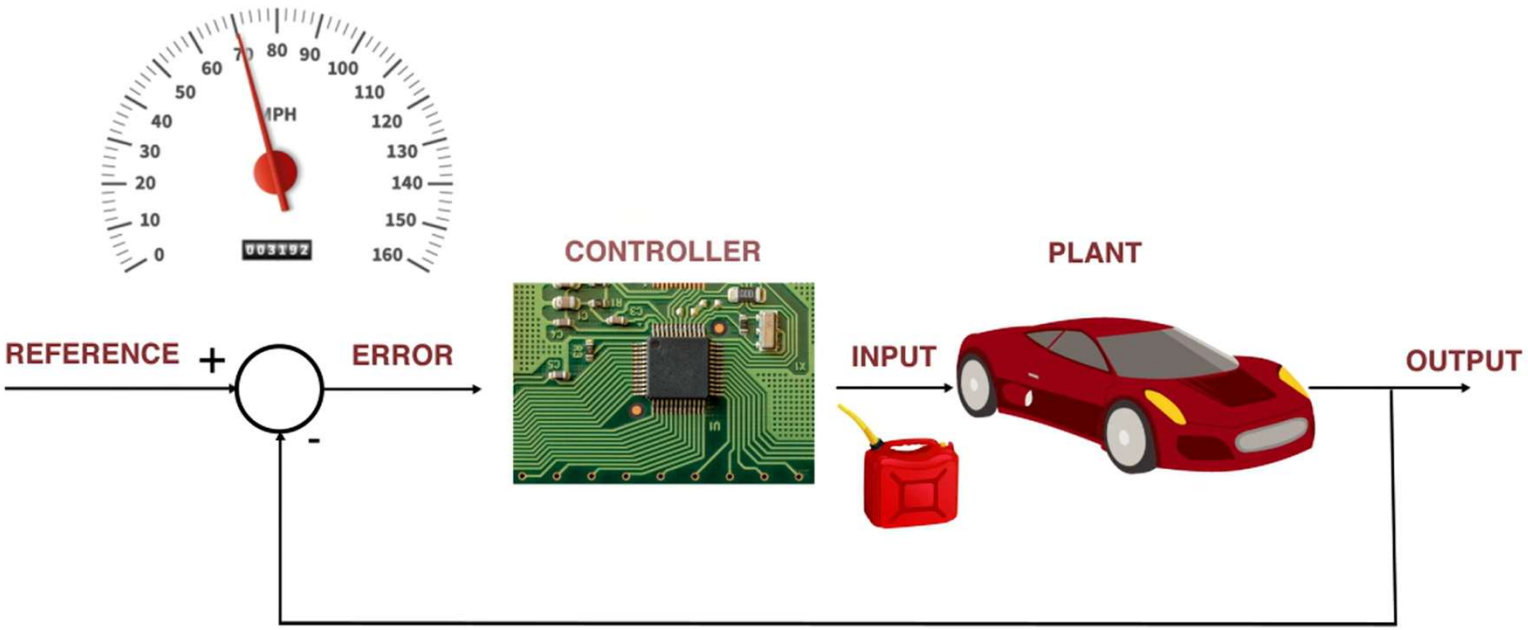
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Example- Cruise Control

Aim: Keep the speed at a set point , say 70 MPH

- Desire speed (Reference) = 70 MPH
- Current speed (current output) = 40 MPH
- Difference (Error)= 30 MPH





- The speed sensor sends the actual speed to the comparison point (summing junction).This closes the loop and making it a closed-loop feedback control system.
- The system continuously compares the *actual speed* with the *set speed* to calculate the *error*.
- The Controller processes the error signal and decides how much fuel or throttle to apply. If the car is too slow, it increases throttle; if too fast, it reduces it.

Closed-Loop Control Examples

Example	Description	Feedback Variable
Room Thermostat (HVAC)	Keeps temperature constant by turning heating/cooling on or off.	Room temperature.
Automatic Voltage Regulator	Maintains constant output voltage in a generator.	Output voltage (voltage sensor)
DC Motor Speed Control	Adjusts motor voltage/current to maintain speed.	Motor speed sensor or tachometer.
Drone Altitude Control	Keeps constant height using sensors.	Altitude via air pressure sensor (barometer) ultrasonic sensor.
Automatic Water Level Controller	Fills tank until a certain level.	Water level sensor.

Various Control Strategies

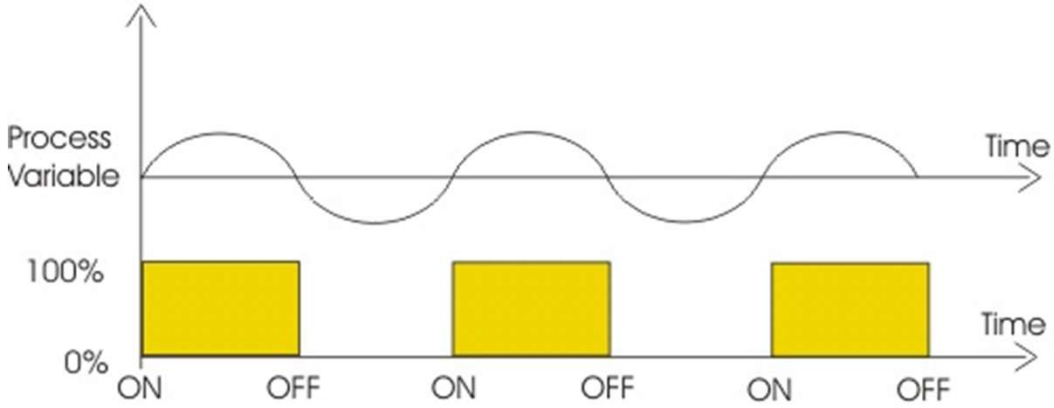
- **On–Off Control (Bang–Bang Control)**
- **Proportional–Integral–Derivative (PID) Control**
  - Proportional (P) Control
  - Proportional–Integral (PI) Control
  - Proportional–Derivative (PD) Control
  - Proportional–Integral–Derivative (PID) Control

- **Adaptive Control:** **Controller parameters change automatically** based on system behavior or environment. (model reference adaptive control, direct, indirect..)
- **Model-Based Control (MPC – Model Predictive Control):** Uses a **mathematical model** of the system to predict future behavior and optimize control actions.
- **Fuzzy Logic Control:** Uses **linguistic rules** (e.g., “If temperature is high, decrease heater slightly”) instead of precise math models.
- **Neural-Network / AI-Based Control:** Uses **machine learning models** to learn system dynamics and optimize control actions.
- ....

Closed-Loop Control Strategies

On–Off Control (Bang–Bang Control):

- Controller switches abruptly between two states.
- Output switches fully ON or OFF depending on whether the measured variable is below or above the setpoint.
- Example: Room thermostat controlling a heater.
- Pros: Simple and low-cost.
- Cons: Causes oscillations around the setpoint.



PID Controller

PID stands for **Proportional-Integral-Derivative**, a widely used feedback control algorithm in industrial applications.

P (Proportional): Responds to the *current error*.

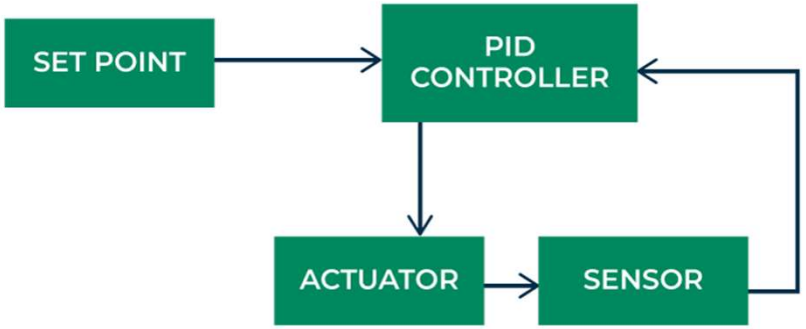
- Bigger error = stronger correction.
- Fast but leaves steady-state error.

I (Integral): Adds up *past errors* over time.

- Eliminates leftover error, reaches exact setpoint.

D (Derivative): Looks at the *rate of change* of error.

- Predicts overshoot, smooths response, reduces oscillation.



Proportional (P) Control

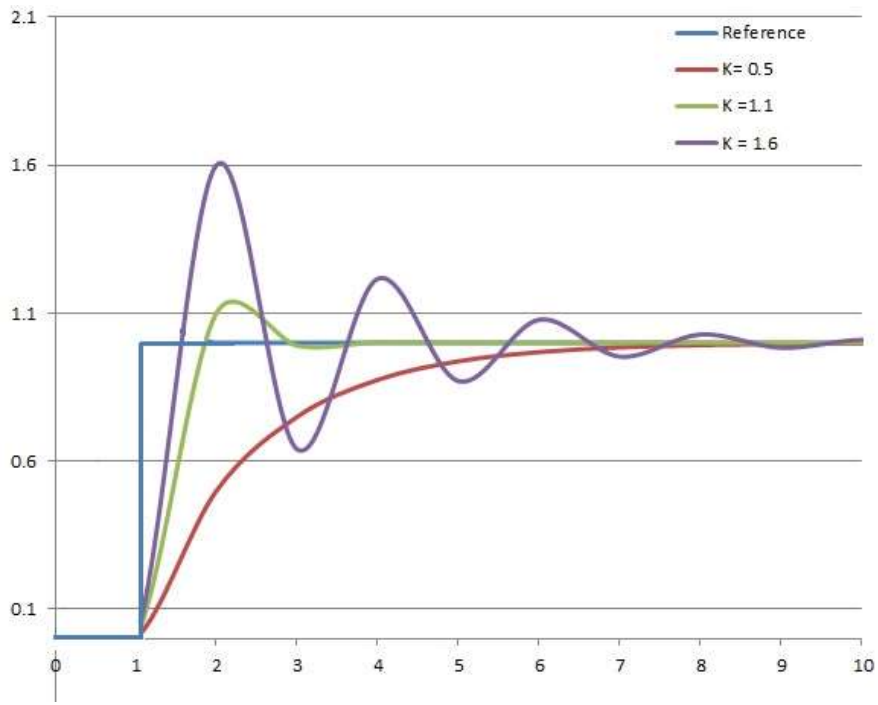
- A **P-controller** is the **simplest** form of feedback control.
- It produces an output **proportional** to the **current error.**
- Proportional control gives smooth and intuitive correction.

$$u(t) = K_p\, e(t)$$

$u(t)$ = controller output (control signal sent to actuator)

$e(t) = r(t) - y(t)$ = error (difference between desired output $r(t)$ and actual output $y(t)$)

K_p = **Proportional gain**, a constant that decides *how strongly* the controller reacts to the error.



- Kp is the proportional gain
- The gain Kp determines how aggressively the system reacts.

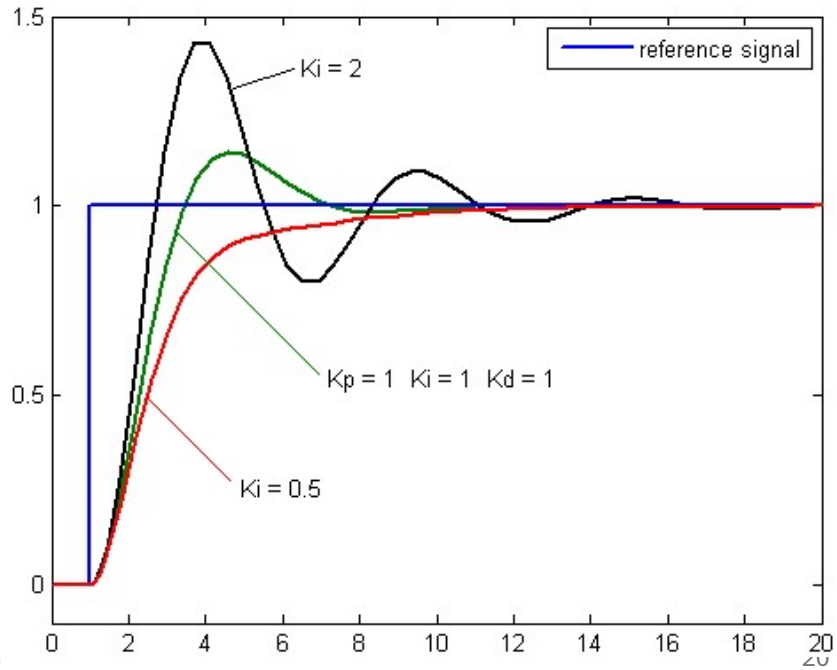
Pros: Smooth control response.
Cons: Steady-state error usually remains.

Integral (I) Control: Fixing the Past Error

- The **integral term** looks at the **accumulated error over time**.
  Even if the proportional term brings the error close to zero, sometimes a **small steady error** remains (e.g., motor running slightly slower than target).
- The integral term continuously sums up all past errors this error and applies correction until it (steady-state-error) disappears.

$$u_I(t) = K_i \int_0^t e(\tau)\, d\tau$$

- Ki = Integral gain



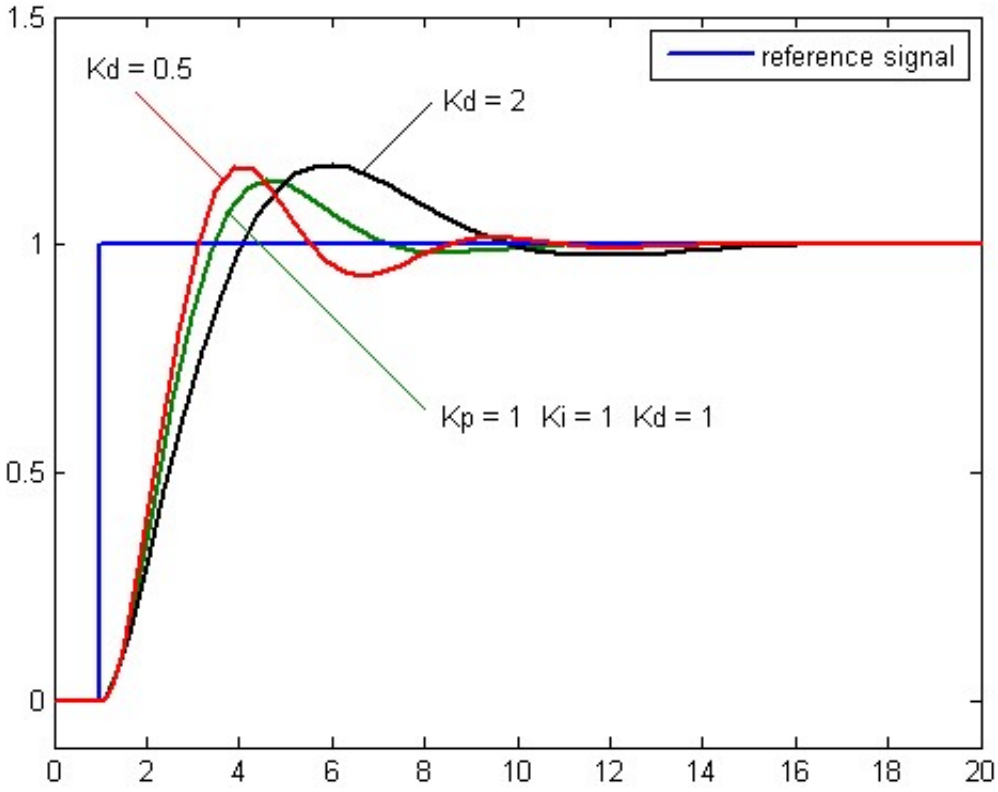
Derivative (D) Control: Predicting the Future Error

- The **derivative term** predicts how the error is changing; it acts on the **rate of change of error.**
- It provides a **damping effect**, reducing overshoot and improving stability.

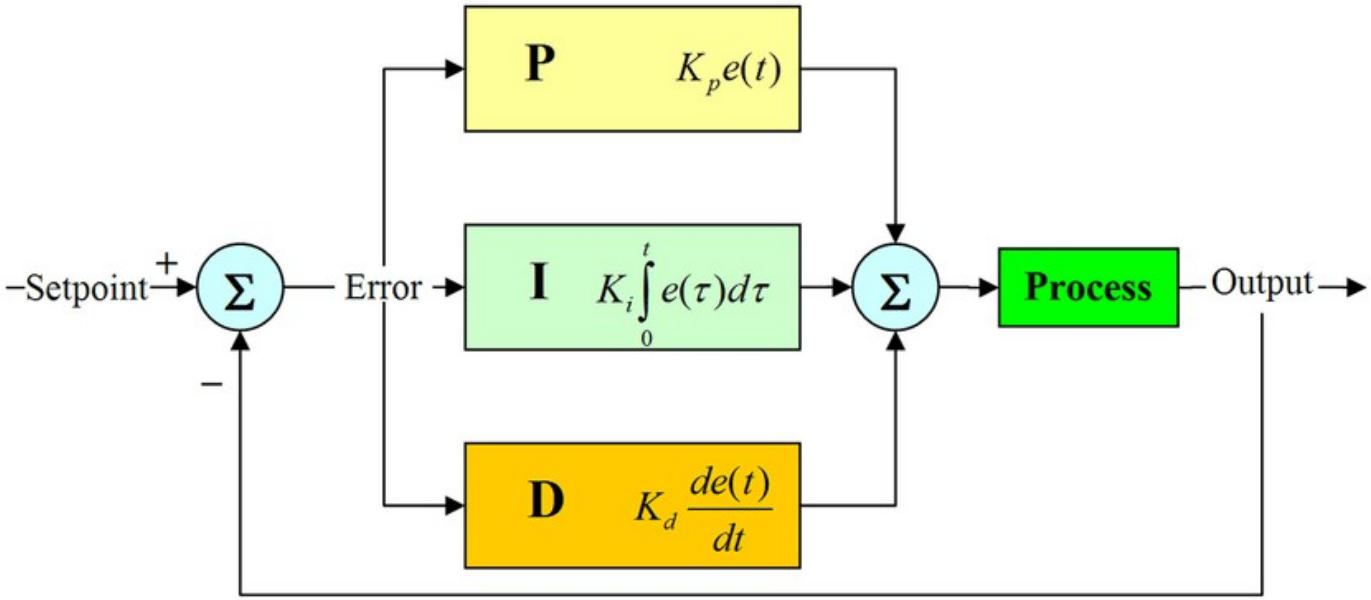
$$u_D(t) = K_d \frac{de(t)}{dt}$$

- Kd = Derivative gain
- It reacts to how fast the error is changing:
- large changes → strong correction.

Downside:
Sensitive to noise. so Kd should be small or filtered.



PID controller Mathematical Expression



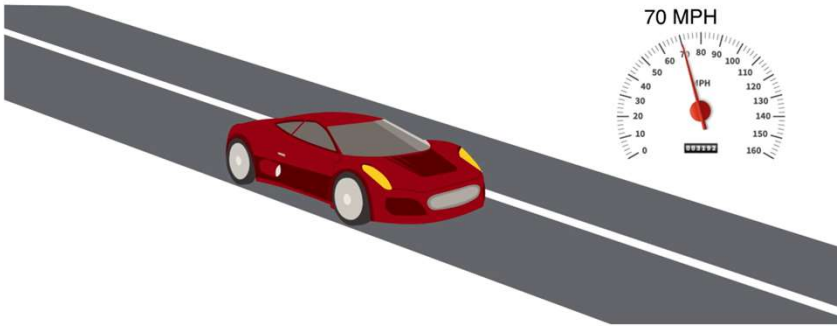
$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt}$$

Example: Car Speed Control (Cruise Control)

The **ECU (Electronic Control Unit)** continuously measures the **actual speed** and compares it with the **setpoint (70 mph)**.

- If the car slows down (e.g., uphill), the ECU **increases throttle** to add more power.
- If the car speeds up (downhill) → the ECU **reduces throttle** to slow it down.

- This automatic adjustment is done by a **PID controller** inside the ECU.
- PID controller does, it **automatically adjusts the "input"** (throttle) to keep the **output** (speed) at the **setpoint** (70 mph).



Cruise Control- PID Control

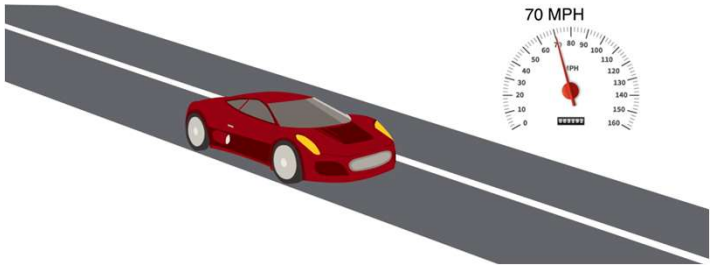
Proportional (P):

- Proportional (P) control produces a correction directly proportional to the current error , the **larger the error, the stronger the response**.
- Reacts to how far the car's speed is from 70 mph, More error → more throttle.

- Setpoint speed is 70 mph.

Integral (I):

- The integral part looks at the history of the error ; how long the car has been above or below the target speed.
- Slowly adds or reduces throttle to eliminate small steady errors.



Derivative (D):

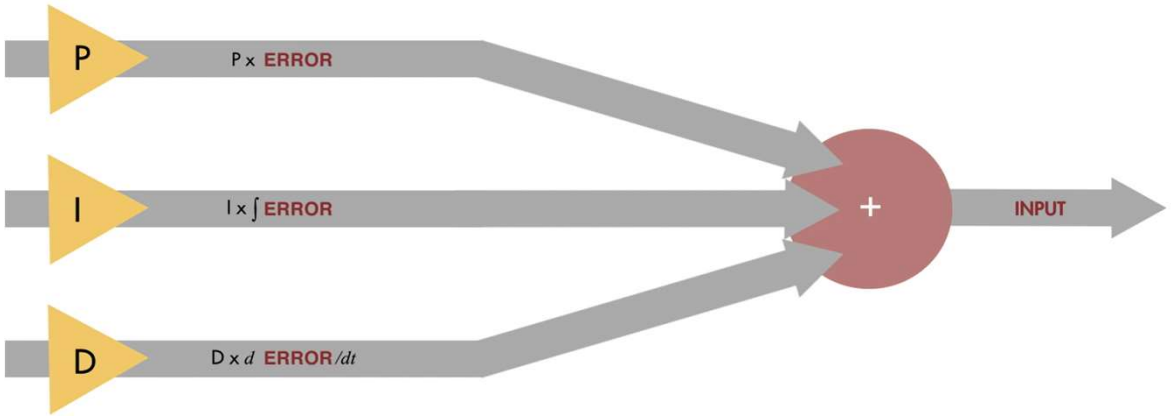
- The derivative part looks at how fast the error is changing , and it predicts the **future trend**.
- If the car is accelerating too fast toward 70 mph, it reduces throttle early to prevent overshoot.

PID Controller

PID combines *present, past,* and *future* error information; responding quickly, correcting accurately, and settling smoothly.

- The proportional (P) component provides immediate correction,
- The integral (I) component addresses sustained errors, and
- The derivative (D) component dampens the response to prevent excessive oscillations.

A PID controller looks at three things:
1. Now – how far from target (Proportional)
2. Past – how long off-target (Integral)
3. Future – where trend is heading (Derivative)



Practical PID –Tuning

Rule of Thumb: Start with P → then I → finally D.

Start with P, then add I to eliminate errors, and finally D for fine-tuning

PID Tuning - Cruise control example

- Far from target speed → press accelerator more (P).
- Been below speed too long → press harder (I).
- Approaching target too fast → ease off (D).